Network Reconnaissance and Port Scanning using Nmap

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Project Type: Cybersecurity Lab Project

Tools Used: Kali Linux, Nmap

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➢ Project Overview:

This project focuses on basic network reconnaissance using Nmap in a Kali Linux environment. The main objective of this project is to understand how network scanning works and how it can be used to discover devices connected to a network.

By using Nmap, it is possible to identify active hosts, open ports, and services running on those ports. This helps in understanding how different systems communicate within a network.

Network scanning is an important part of cybersecurity because it helps security professionals analyze the structure of a network and identify potential security risks. In the early stages of security testing, reconnaissance techniques are used to gather useful information about systems and services.

In this project, Nmap was used in Kali Linux to perform basic scans on a local network. The results of these scans help demonstrate how information about devices and services can be discovered during the reconnaissance phase of cybersecurity testing.

➢ Tools Used:

1. Kali Linux:

Kali Linux is a Linux distribution that is specially designed for penetration testing and cybersecurity tasks. It is widely used by security professionals and ethical hackers to test and analyze the security of computer systems and networks. Kali Linux comes with many pre-installed tools that help in

activities such as network scanning, vulnerability analysis, password testing, and digital forensics.

One of the main advantages of Kali Linux is that it provides a large collection of security tools in a single operating system. Tools like Nmap, Metasploit, and Wireshark are commonly used for different types of security testing. Because of these tools, Kali Linux is very useful for learning and practicing cybersecurity techniques.In this project, Kali Linux was used as the main operating system to perform network reconnaissance using Nmap. It provides a simple and effective environment to run scanning tools and analyze the results during the initial phase of security testing.

2. Nmap:

Nmap (Network Mapper) is a popular open-source tool used for network discovery and security auditing. It is commonly used by cybersecurity professionals and network administrators to scan networks and gather information about devices connected to them.

Nmap helps in identifying active hosts, open ports, and services running on those ports. By performing different types of scans, it is possible to understand how a network is structured and which services are available on a system. This information is very useful during the reconnaissance phase of security testing.

Nmap is included by default in Kali Linux, which makes it easy for security learners and professionals to perform network scans. It supports many scanning techniques such as host discovery, port scanning, service detection, and operating system detection.

In this project, Nmap was used to perform basic scans on a local network to discover active devices and check which ports and services were available on them.

- **Host Discovery**

Host discovery is the process of finding active devices connected to a network. It is one of the first steps in network reconnaissance because it helps identify which systems are currently online and responding in the network.

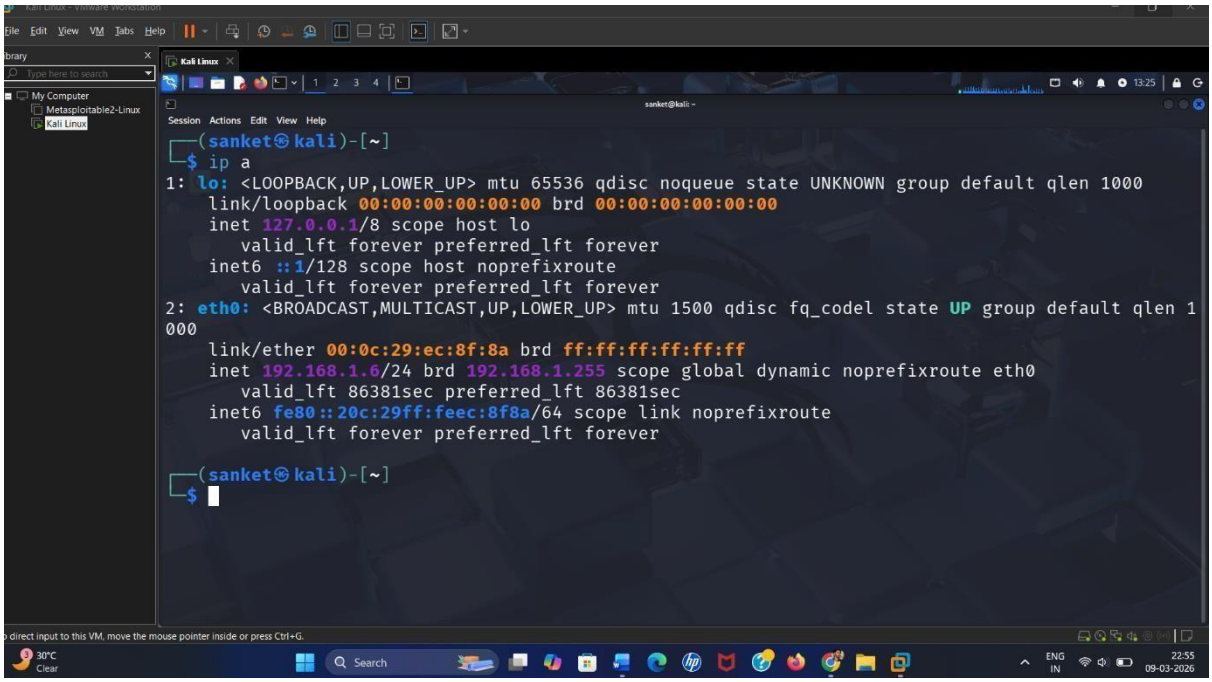
By performing host discovery, a security tester can detect the IP addresses of active hosts before starting deeper scanning. This step helps in saving time because scans can be focused only on the systems that are active. Tools like Nmap provide different techniques for host discovery, such as sending ICMP echo requests, TCP packets, or ARP requests to check if a device responds. If the device replies, it means that the host is active on the network.

In this project, host discovery was performed using Nmap on Kali Linux to identify active devices in the local network. The results helped in understanding which systems were available for further scanning.

- **Command:** ip a

Before starting host discovery, the network interface information was checked using the ip a command in Kali Linux. This command shows details about the system's network interfaces and the assigned IP address. From the output, it was observed that the system IP address was 192.168.1.6 with a /24 subnet. Based on this information, the network range 192.168.1.0/24 was selected for scanning. After identifying the

network range, Nmap was used to perform host discovery using the command nmap -sn 192.168.1.0/24 to identify active hosts in the local network.



- **Command:** nmap -sn 192.168.1.0/24

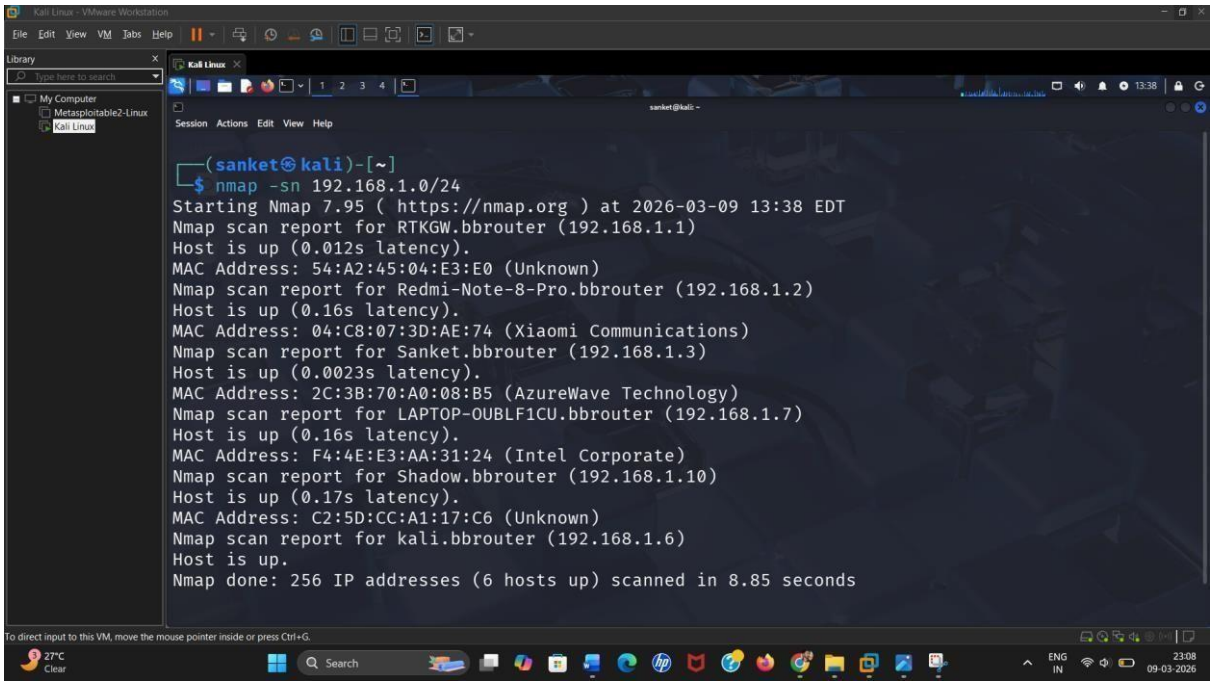
The command nmap -sn 192.168.1.0/24 is used to perform host discovery on a network. This command checks which devices are active in the specified network range.

In this command, Nmap sends packets to all IP addresses in the network 192.168.1.0/24 to see which systems respond. If a system replies, it means that the host is active and connected to the network.

The -sn option tells Nmap to perform a ping scan only, which means it will detect active hosts but will not perform port scanning. This makes the scan faster because it only focuses on identifying live devices.

In this project, the command nmap -sn 192.168.1.0/24 was used on Kali Linux to discover active hosts in the local network before performing further scanning.

- **Command:** nmap -sn 192.168.1.0/24



- **Host Discovery Result:**

After running the command nmap -sn 192.168.1.0/24 using Nmap in Kali Linux, the scan showed the active devices connected to the local network. From the output, several devices were detected such as the router, a mobile device, laptops, and the Kali Linux system used to perform the scan. Each device was displayed with its IP address and MAC address. The output also showed the latency value, which indicates how quickly each device responded to the scan.

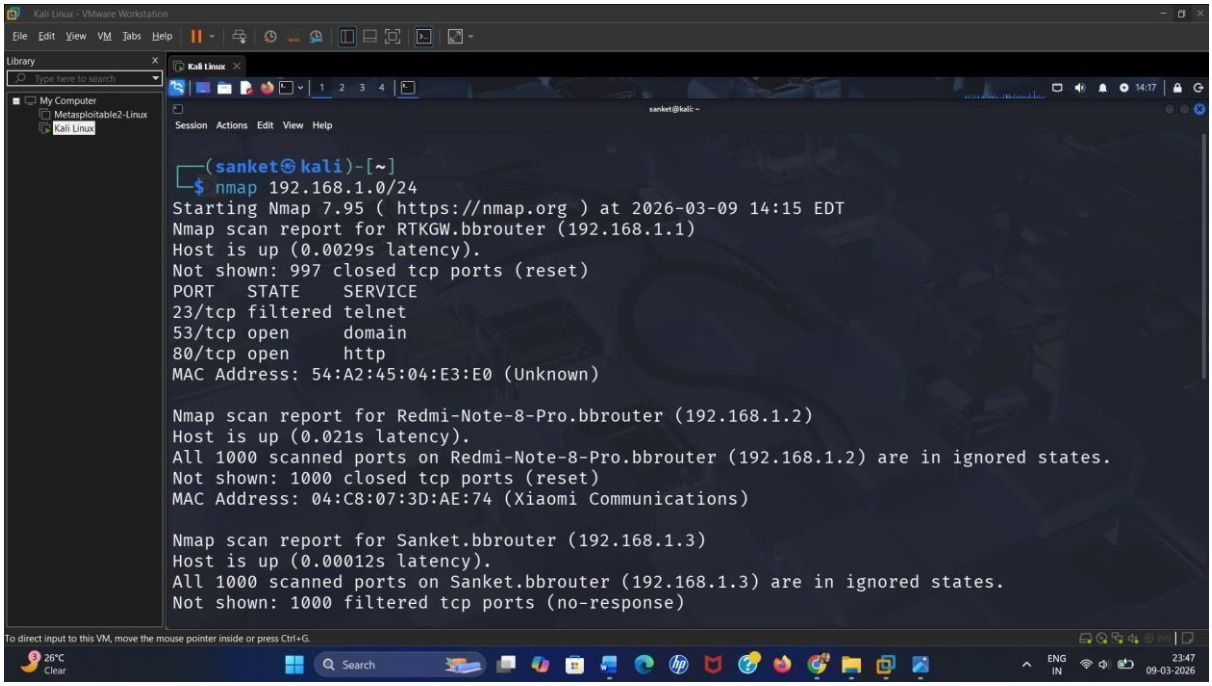
At the end of the scan, it was observed that a total of 256 IP addresses were scanned and 6 hosts were found active on the network. This information helps in identifying which devices are currently available on the network and can be used for further analysis.

➢ Port Scanning:

Port scanning is used to find open ports on a target system. Open ports show which services are running on a device and how the system communicates over the network. In this project, port scanning was performed using Nmap in Kali Linux. The scan helps in identifying open ports on the target machine and gives an idea about the services that may be running on those ports.

This information is useful during network analysis because it helps understand the services available on a system and can also help identify possible security risks.

➢ Command: nmap 192.168.1.0/24



➢ Port Scanning Result:

After running the command nmap 192.168.1.0/24 using Nmap in Kali Linux, the scan showed the status of ports on the active devices in the local network.

From the scan results, it was observed that the router with IP address 192.168.1.1 had some open ports. For example, port 53 (domain) and port 80 (http) were open, which means DNS and web services may be running on that device. Port 23 (telnet) was shown as filtered, which means the system did not respond to the scan request for that port.

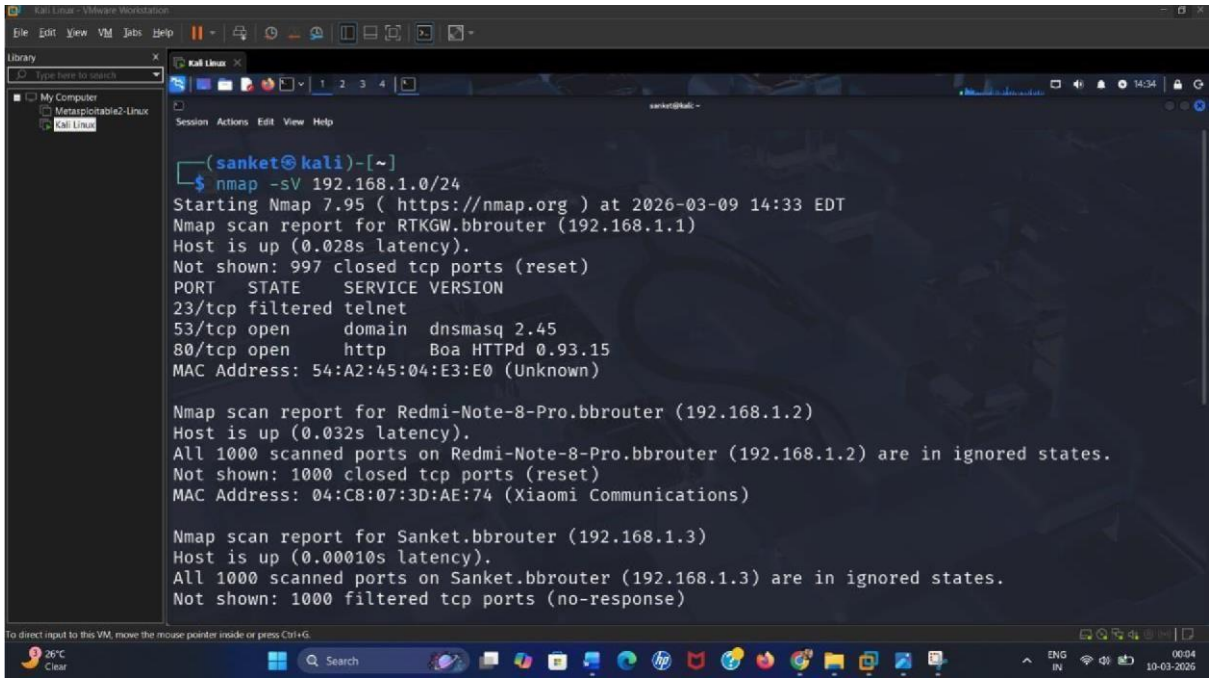
For the other devices in the network, most of the scanned ports were closed or filtered. This means that no common services were detected on those ports during the scan.

Overall, the port scanning process helped in identifying open ports and understanding which services might be running on devices in the local network.

➢ Service Detection:

Service version detection is used to find the exact version of the service running on an open port. When a port is open, it usually means a service like a web server, DNS server, or SSH is running on that port. By using service detection, we can identify which software and version is being used. This information is useful in cybersecurity because some versions of services may have known vulnerabilities. In this project, Nmap was used with the -sV command to detect the service versions running on the devices in the network. This helped us understand more details about the services available on each open port.

- **Command:** nmap -sV 192.168.1.0/24



- **Service Version Detection Result:**

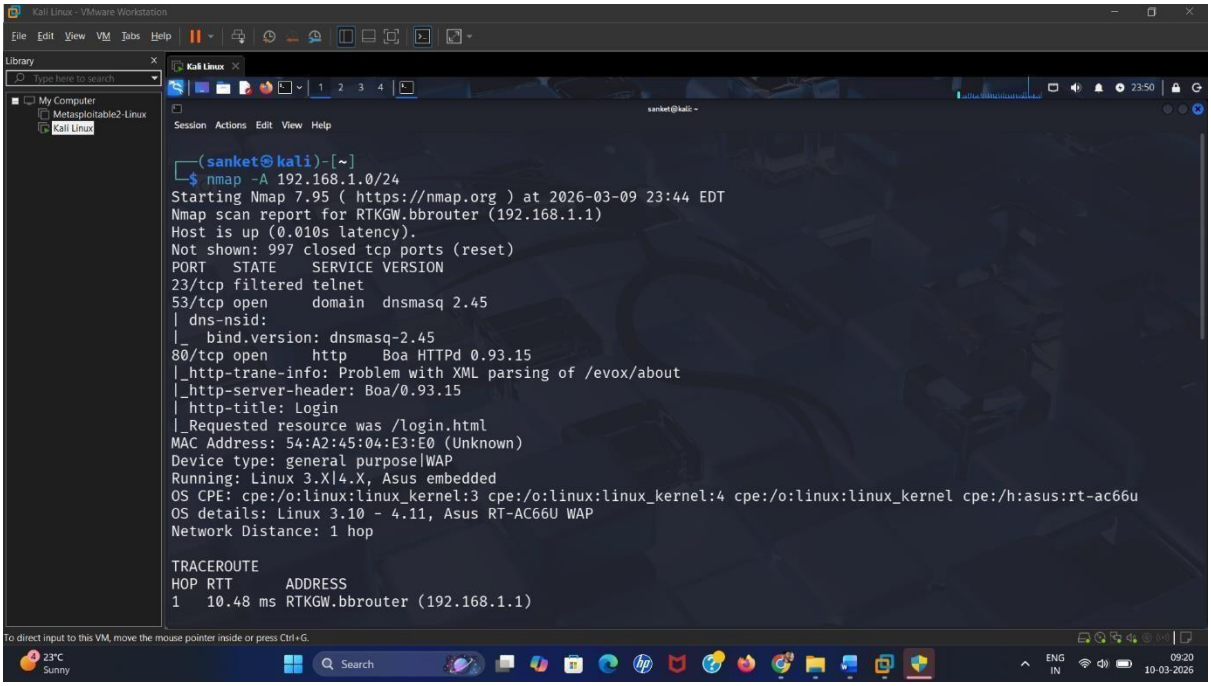
  After running the command nmap -sV 192.168.1.0/24, Nmap scanned the devices available in the local network and tried to detect the services and their versions running on open ports. From the scan results, the router with IP address 192.168.1.1 showed some open ports. Port 53 was open and running the DNS service (dnsmasq 2.45) which helps in resolving domain names in the network. Port 80 was also open and running Boa HTTPd 0.93.15, which is usually used for the router's web management page. Port 23 (Telnet) was shown as filtered, which means the port might be protected or blocked by a firewall.

- **Aggressive Scan:**

The -A option in Nmap enables advanced scanning features. It performs multiple types of scans at the same time, such as operating system detection, service version detection, script scanning, and traceroute. This type of scan

helps in getting deeper information about the target devices, including the services running on them and possible details about the operating system. Aggressive scanning is useful during network reconnaissance because it provides more detailed results compared to normal scans. However, it generates more network traffic and may be detected by security monitoring systems, so it should only be used in authorized environments like a lab or testing network.

- **Command:** nmap -A 192.168.1.0/24



- **Aggressive Scan Result:**

Using the command nmap -A 192.168.1.0/24 in Kali Linux, the scan found active devices on the network and their running services. The router at 192.168.1.1 was active. Most ports were closed, but some were open or filtered. Port 53 was open running DNS (dnsmasq 2.45), and port 80 was open with a web server (Boa HTTPd 0.93.15) showing a login page at /login.html. Port 23 (Telnet) was filtered. The router is a Linux-based Asus device with kernel version 3.10 – 4.11, and it is directly connected to the

network (1 hop). The scan successfully showed the active device, open ports, and basic service information.

- **Summary of Nmap Scan Results:**

IP Address	Port	Service	Status
192.168.1.1	53	DNS	Open
192.168.1.1	80	HTTP	Open
192.168.1.1	23	Telnet	Filtered

- **Conclusion:**

This project demonstrated how network reconnaissance and port scanning can be performed using Nmap in Kali Linux. Different scanning techniques such as host discovery, port scanning, service detection, and aggressive scanning were used to analyze the local network. The scans helped identify active devices, open ports, and services running on them. Through this project, practical knowledge of network scanning and the importance of reconnaissance in cybersecurity was gained.

➢ References

1. Nmap Official Documentation – https://nmap.org

2. Kali Linux Official Website – https://www.kali.org

3. Cybersecurity Fundamentals – Network Scanning Concepts